Name: Lee Foster III

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CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, CLOSED BOOK, TIME:120 MIN

1. Ethylene oxide C_2H_4O is produced isothermally at 300 °C by the vapor-phase catalytic oxidation of ethylene C_2H_4 with air.

19/40

$$2C_2H_4 + O_2 \rightarrow 2C_2H_4O$$

$$r'_{C_2H_4} = -k\ P_{C_2H_4}^{1/3}\ P_{O_2}^{2/3} \qquad k = 4 \times 10^{-8} \frac{mol}{Pa.kg.s}\ \ at\ 300°C$$

$\frac{mol}{Pa\,kg\,s}(Pa)$

$r' = \left[\frac{mol}{kg\,s}\right]$

The process takes place in two sequential fluidized bed reactors. The feed contains ethylene, oxygen, and nitrogen (inert) with molar flow rates of 2, 2, and 6 $\frac{mol}{s}$. The pressure of the feed, the first reactor, and the second reactor are maintained at 10, 6, and 4 bar respectively ($1\ bar = 10^5\ Pa$).

Estimate the catalyst weight in each reactor, given that the ethylene conversion is 50% after the first reactor and 80% after the second.

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2. The elementary liquid-phase reactions

$$(1) \qquad A + 2B \rightarrow 2C$$
$$(2) \qquad A + C \rightarrow 2D$$

are carried out in a 50 dm^3 well-mixed CSTR equipped with a heating jacket. A mixture of A and B is fed to the reactor at the volumetric flow rate of $1 \frac{dm^3}{s}$, temperature of $500K$ and concentrations $C_{A0} = 1.0 \frac{mol}{dm^3}$ and $C_{B0} = 2.0 \frac{mol}{dm^3}$. At steady state, the outlet temperature is measured to be $450K$ and the outlet concentrations are $C_A = 0.25 \frac{mol}{dm^3}$, $C_B = 1.0 \frac{mol}{dm^3}$, $C_C = 0.75 \frac{mol}{dm^3}$, and $C_D = 0.5 \frac{mol}{dm^3}$.

Additional information:

Heat capacities: $C_{PA} = 20 \frac{J}{mol.K}$, $C_{PB} = 20 \frac{J}{mol.K}$, $C_{PC} = 30 \frac{J}{mol.K}$, $C_{PD} = 25 \frac{J}{mol.K}$

Heats of reactions: $\Delta_{r1}H = 20 \frac{kJ}{mol\ A}$ and $\Delta_{r2}H = -10 \frac{kJ}{mol\ A}$

Heating fluid enters the jacket at a temperature $T_{a0} = 600\ K$, mass flow rate: $\dot{m}_a = 0.02 \frac{kg}{s}$ with heat capacity $C_{Pa} = 4500 \frac{J}{K.kg}$

10/20 2.1. Write the mole balance equations for all four species and estimate the reaction rates $r_{A(1)}$ and $r_{A(2)}$ within the reactor.

10/20 2.2. Write the energy balance equation for the reactor and estimate the rate of heat transfer into the reactor $\dot{Q}$, assuming negligible shaft work.

5/20 2.3. Write down the energy balance for the heating fluid and calculate the exit temperature of the heating fluid. Estimate the overall heat transfer coefficient, given the heat transfer area is $A = 120\ dm^2$.

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1.)

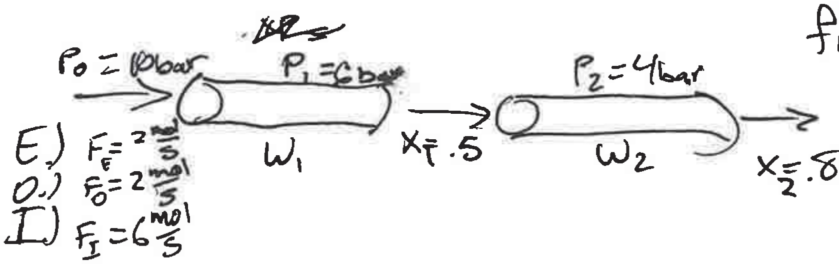
$E = C_2H_4$

$r'_{C_2H_4} = -K P_{C_2H_4}^{1/3} P_{O_2}^{2/3}$

$K = 4\times10^{-8} \frac{mol}{Pa\ Kg\ s}$

$T = 300°C$

find W_1 \& W_2



$P_{tot} = 6\ bar = 6\times10^5\ pa$

$$\frac{dF_E}{dW} = r'_{C_2H_4}$$

reactor 1

$P_{C_2H_4} = \left(\frac{MW_E}{MW_{C_2H_4}}\right)(P_{tot}) =$

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$= (.19)(6\times10^5) = 1.14\times10^5\ Pa$

$P_{O_2} = (.22)(6\times10^5) = 1.32\times10^5\ pa$

$$\frac{F_{E0} - F_E}{W_1} = r'_{C_2H_4}$$

reactor 2

$P_{C_2H_4} = (.19)(4\times10^5) = 7.6\times10^4\ pa$

$P_{O_2} = (.22)(4\times10^5) = 8.8\times10^4\ pa$

$$\frac{F_{E0} - F_{E0}X_1}{W_1} = r'$$

$$W_1 = \frac{F_{E0}(1-X_1)}{r'} = \frac{2\frac{mol}{s}(1-.5)}{4\times10^{-8}\frac{mol}{Pa\,Kg\,s}(1.14\times10^5)^{1/3}(1.32\times10^5)^{2/3}} = \frac{1}{.005} = 200\,Kg$$

w_1

$$W_2 = \frac{F_{E_1} - F_{E_2}}{r'} = \cancel{}$$

$$W_2 = \frac{F_{E_1}(1-X_2)}{r'} = \frac{1(1-.8)}{4\times10^{-8}(7.6\times10^4 pa)^{1/3}(8.8\times10^4 pa)^{2/3}} = \frac{.2}{.0034}$$

$W_2 = 57.7\ Kg$

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2.)

$V = 50 dm^3$ $\quad C_{A0} = 1.0 \frac{mol}{dm^3}$ $\quad$ S.S

$F_{a0} = 1 \frac{dm^3}{S}$ $\quad C_{B0} = 2 \frac{mol}{dm^3}$

$T_0 = 500K$ $\quad T_F = 450K$

$2r_{A1} = r_{B1} =$

$r_{A1} = \frac{1}{2} r_{B1}$

S.S

in - out + gen - cons = 0

2.1)

A.) $F_{A0} - F_A - \cancel{\tfrac{1}{2}}(r_{A(1} + r_{A2}) = 0$

B.) $F_{B0} - F_B \cancel{-2}(r_{A1}) = 0$

C.) $F_{C0} - F_C + 2r(\cancel{A_1}) - r(A_2) = 0$

D.) $F_{D0} - F_D + 2r(A_1) = 0$

$r_{A1} = \frac{1}{2}(F_{B0} - F_B) = (\frac{1}{2})(1 \cdot 2 - 1 \cdot 1)$

$= \frac{1}{2}$ or $.5 \frac{mol}{s}$

$r_{A2} = r_{A1} + F_A - F_{A0}$

$= .5 + (1 \cdot 1 - 1 \cdot .25)$

$= 1.25 \frac{mol}{s}$

2.2.)

$$\frac{dU^{0}}{dt} = \sum_\alpha F_\alpha h_\alpha(T_0) - \sum_\alpha F_\alpha h_\alpha(T) - \sum_\alpha F_\alpha \nu_\alpha h_\alpha(T) X + \dot{Q} + \dot{W}_s^{0}$$

$$0 = \sum_\alpha F_\alpha (h_\alpha(T_0 - T)) - [\Delta H_{rxn}] F_A X + \dot{Q}$$

$$0 = \sum_\alpha F_\alpha C_{p_A}(T_0 - T) - [H^o_{rxn}(ref) + \Delta C_p(T_{ref})] F_A X + \dot{Q}$$

$$\dot{Q} = [H^o_{rxn}(ref) + \Delta C_p(T_{ref})] F_A X - \sum_\alpha F_\alpha C_p(T_0 - T)$$

$$= (10 \frac{kJ}{mol A}) + 95 (\frac{J}{mol \cdot K})(1) - (3 (\cancel{4500}\ 20)(100K))$$

$$\dot{Q} = 10.8 \frac{kJ}{S}$$

2.3

$$\dot{Q}_{rxn} = \dot{Q}_{trans} = h \dot{m}\, C_{p_a} = (.2 \frac{kg}{s})(4500 \frac{J}{K \cdot kg})(h)$$

$$\frac{1000}{1}(10.8) = h(900 \frac{J}{K \cdot s}) \quad h = 12 \frac{1}{K}$$

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